

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

**RESPONDENTS' NOTICE OF PRIOR ART AS TO
U.S. PATENT NOS. 10,642,359, 10,231,675, 10,945,677, AND 10,978,789**

In accordance with the Procedural Schedule (Order No. 8) and Ground Rule No. 8, Respondents Ouraring Inc. and Oura Health Oy (collectively, "Oura" or "Respondents") hereby provide their Notice of Prior Art. Respondents provide the required information in the attached Exhibit A. Respondents expressly reserve the right to rely on these references, either singly or in any combination, to establish the invalidity of (1) U.S. Patent Nos. 10,642,359 ("the '359 Patent"); (2) 10,231,675 ("the '675 Patent"); (3) 10,945,677 ("the '677 Patent"); and (4) 10,978,789 ("the '789 Patent") (collectively "the Asserted Patents"), or to demonstrate the relevant state of the art and/or level of skill in the art.

Accordingly, Respondents may rely on each prior art reference set forth in the attached Exhibit A as anticipatory and/or as rendering obvious alone or in combination with any other prior art one or more of the asserted claims of the Asserted Patents. Each listed patent, publication, product, prior art, or other documentation is effective as prior art at least as early as the corresponding date indicated and prior to the priority date of the Asserted Patents.

In addition to the references, products, and persons listed in Exhibit A to this Notice, Respondents hereby give notice that they may rely on, and therefore incorporate by reference: (i)

prior art contained or referenced in the prosecution history of the Asserted Patents, in the Asserted Patents, or in any patent, application, or prosecution history that is related to the Asserted Patents; (ii) prior art identified or produced by Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (collectively, “Samsung” or “Complainants”) or Subpoenaed third party in this Investigation; (iii) prior art admitted into evidence or identified as an exhibit at the hearing or deposition; (iv) prior art identified during the remainder of discovery; (v) prior art cited or referenced in any of the references identified in Exhibit A to this Notice (regardless of whether a cited reference appears separately in Exhibit A); (vi) prior art cited in any expert report served during this Investigation; or (vii) prior art cited or referenced in the file histories for all of the patents and patent applications listed in Exhibit A or identified in the prior categories (i) – (vi), which became publicly available as of the dates of the patents issued, or the dates the patent applications published, whichever is applicable. Respondents reserve the right to rely on documents and information that are not prior art, and therefore need not be disclosed in this Notice, for other purposes including, but not limited to, evidence of inherency in any of the prior art or establishing the prior art as prior art (*e.g.*, documents establishing the sales dates of prior art products), or establishing the knowledge of one of ordinary skill in the art or how one of ordinary skill in the art would interpret any disclosure in any prior art reference or asserted patents.

Respondents reserve the right to rely upon any person identified as an author or inventor of any prior art identified herein as a prior inventor or as having prior knowledge of or as having previously used or offered for sale the alleged inventions of the Asserted Patents. Respondents may also rely on individuals, the names and addresses of which are within Complainants’ possession, including Complainants’ employees, former employees, consultants, and/or other

persons affiliated with Complainants who may have knowledge of, or have previously used or offered for sale, the alleged inventions of the Asserted Patents.

Additionally, Respondents reserve the right to supplement or amend this notice as discovery proceeds, including third party discovery. Respondents note that discovery, including third party discovery, is ongoing and fact discovery does not close until May 29, 2026. To date, Complainants have only provided preliminary infringement and DI contentions and claim construction is ongoing, which limits Respondents' ability to understand the scope of potentially relevant prior art. In addition, expert reports have not been exchanged, and expert discovery has not commenced. Respondents reserve the right to amend and/or supplement this Notice of Prior Art and the attached Exhibit A upon written notice for good cause shown to identify prior art references, products, and persons that may be discovered after the date of this Notice or to identify additional prior art that may be rendered material based upon positions Complainants or their expert witnesses may adopt in this Investigation, or if Complainants rely on any claims, for any purpose, that have not been identified to date. Respondents also reserve the right to amend this Notice should it be discovered that a reference cited during prosecution of an asserted patent (or a foreign counterpart or family member of an Asserted Patent) was omitted from the following list through inadvertence.

Respondents' Notice of Prior Art is based on the priority dates set forth in Respondents' initial invalidity contentions. *See* Respondents' Invalidity and Ineligibility Contentions for U.S. Patent Nos. 10,642,359, 10,231,675, 10,945,677, and 10,978,789 (served on March 16, 2026). Complainants have not met their burden of showing on a claim-by-claim basis that it is entitled to a priority date earlier than any of those dates.

Respondents may assert defenses that, if successful, could potentially result in one or more of the asserted claims not being entitled to the priority dates asserted by Complainants. In the event

one or more of the asserted claims is not entitled to such priority date, Respondents reserve the right to rely on intervening prior art including other related patents and published applications, as well as prior art not listed in Exhibit A.

Each patent, patent application publication, non-patent publication, and product listed in Exhibit A includes and incorporates all references cited therein, regardless of whether a cited reference appears separately in Exhibit A.

Respondents may also rely on non-prior-art grounds of invalidity for and/or unenforceability of the Asserted Patents. Respondents incorporate by reference herein their interrogatory responses, contentions, and defenses concerning these grounds of unenforceability and invalidity, including, without limitation, Respondents' Joint Disclosure of Initial Invalidity Contentions (and any amendments or supplements thereto) and Respondents' briefing on claim construction.

Dated: May 13, 2026

Respectfully submitted,

/s/ Jasjit S. Vidwan

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Exhibit A

EXHIBIT A

I. Patents and Patent Publications

The following table lists prior art patents and patent publications which are identified by patent or patent publication numbers, country or origin, issue/publication date, and priority/filing date. Each listed patent or patent publication became prior art at least as early as the dates provided on the face of the document. Respondents reserve the right to rely upon earlier dates of publication or public availability to the extent such information is uncovered during discovery. Where an asserted patent is identified in relation to a prior art reference, such relationship is exemplary only and does not limit the use of that prior art reference to support the invalidity of any of the asserted patents or to support the state of the art regarding each of the asserted patents.

Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
U.S. Patent App. Pub. 2012/0078066 (“Ahmed”)	Method and Apparatus for Robust Heart Rate Sensing	United States	2012-03-29	2011-12-02	U.S. 10,642,359
Japanese Patent App. Pub. No. 2009/195589 (“Kuroda I”)	Biological Information Processing Apparatus and its Controlling Method	Japan	2009-09-03	2008-02-25	U.S. 10,642,359
U.S. Patent App. Pub. No. 2008/0139952 (“Kuroda II”)	Biological Information Processing Apparatus and its Control Method	United States	2008-06-12	2007-12-06	U.S. 10,642,359
U.S. Patent No. 9,596,999 (“Moon”)	Body-worn Pulse Oximeter	United States	2017-03-21	2009-09-14	U.S. 10,642,359
WO 2012/156427 A1 (“Muzet”)	System and Method for Determining Sleep and Sleep Stages of a Person	France	2012-11-22	2012-05-15	U.S. 10,642,359
U.S. Patent App. Pub. 2008/0234785 (“Nakayama”)	Sleep Controlling Apparatus and Method, and Computer Program	United States	2008-09-25	2007-09-13	U.S. 10,642,359

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
	Product Thereof				
U.S. Patent No. 7,507,207 (“Sakai”)	Portable Biological Information Monitor Apparatus and Information Management Apparatus	United States	2009-03-24	2004-10-05	U.S. 10,642,359
U.S. Patent No. 7,351,206 (“Suzuki I”)	Apparatus for and Method of Biotic Sleep State Determining	United States	2008-04-01	2005-01-31	U.S. 10,642,359
U.S. Patent App. Pub. No. 2015/0245777 (“Torre”)	Detection of Emotional States	United States	2015-09-03	2015-04-20	U.S. 10,642,359
U.S. Provisional App. No. 61/716,405 (“Torre Provisional”)	Wearable Device with an Optical Sensor	United States	N/A	2012-10-19	U.S. 10,642,359
U.S. Patent No. 8,954,135 (“Yuen”)	Portable Biometric Monitoring Devices and Methods of Operating Same	United States	2015-02-10	2013-06-24	U.S. 10,642,359
U.S. Patent No. 8,140,143 (“Picard”)	Washable Wearable Biosensor	United States	2012-03-20	2009-04-16	U.S. 10,642,359
U.S. Patent App. Pub. 2009/0204011 (“Suzuki II”)	Pulse Wave Period Measuring Apparatus, Pulse Wave Period Measuring Method, Heartbeat Period Measuring Method, and Recording	United States	2009-08-13	2008-09-17	U.S. 10,642,359

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
	Medium				
U.S. Patent App. Pub. 2012/0229248 (“Parshionikar”)	Multipurpose Controller for Electronic Devices, Facial Expressions Management and Drowsiness Detection	United States	2012-09-20	2012-03-12	U.S. 10,642,359
U.S. Patent App. Pub. 2011/0132378 (“Levendowski”)	Systems and Methods for Controlling Position	United States	2011-06-09	2010-06-04	U.S. 10,642,359
Korean Patent Application Publication No. 10-2014-0147557 (“Seo”)	Portable Terminal and Method for Controlling Function by Detecting Gesture	Korea	2014-12-30	2013-06-30	U.S. 10,642,359
U.S. Patent No. 10,264,982 (“Ahmed”)	Physiological Measurement System with Motion Sensing	United States	2019-04-23	2014-07-09	U.S. 10,231,675
U.S. Patent No. 8,948,832 (“Hong”)	Wearable Heart Rate Monitor	United States	2015-02-03	2014-05-30	U.S. 10,231,675 U.S. 10,945,677
U.S. Patent No. 6,879,850 (“Kimball”)	Pulse Oximeter with Motion Detection	United States	2005-04-12	2002-03-15	U.S. 10,231,675
WO Pub. No. 2011/024425A1 (“Nishimiya”)	Organism Information Detection Device and Motion Sensor	Japan	2011-03-03	2010-08-23	U.S. 10,231,675
U.S. Patent Pub. No. 2015/0186609 (“Utter II”)	Data Capable Strapband for Sleep Monitoring, Coaching, and	United States	2015-07-02	2014-02-28	U.S. 10,231,675

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
	Avoidance				
U.S. Patent No. 9,808,185 (“Arnold”)	Movement Measure Generation in a Wearable Electronic Device	United States	2017-11-07	2015-09-19	U.S. 10,231,675
U.S. Patent Publication No. 2015/0362977 (“Doniwa”)	Electronic Device, Method, and Computer Program Product	United States	2015-12-17	2014-10-08	U.S. 10,231,675
JP 2013/150847 (“Kamitaki”)	Biological Information Processing Methods and Biological Information Processing Apparatus	Japan	2013-08-08	2013-04-02	U.S. 10,231,675
U.S. Patent Pub. No. 2012/0123226 (“Schwenk”)	Method For Operating a Monitoring System	United States	2012-05-17	2010-07-13	U.S. 10,231,675
U.S. Patent Publication 2014/0206954 (“Yuen”)	Portable Monitoring Devices for Processing Applications and Processing Analysis of Physiological Conditions of a User Associated With the Portable Monitoring Device	United States	2014-07-24	2014-03-20	U.S. 10,231,675
U.S. Patent No. 9,332,940 (“Bologna”)	Compact Wearable Biological Sensor Modules	United States	2016-05-10	2015-01-05	U.S. 10,945,677
U.S. Patent App.	Intelligent Sampling	United States	2015-12-31	2014-06-26	U.S. 10,945,677

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
Pub. No. 2015/0374310 ("Lee")	Of Heart Rate				
U.S. Patent App. Pub. No. 2015/173686 ("Furuta")	Biological Information Measuring Device and Control Method for Biological Information Measuring Device	United States	2015-06-25	2014-12-22	U.S. 10,945,677
U.S. Patent App. Pub. No. 2015/220109 ("von Badinski")	Wearable Computing Device	United States	2015-08-06	2014-11-28	U.S. 10,945,677
U.S. Patent App. Pub. No. 2014/0073486 ("Ahmed")	Systems, Devices and Methods for Continuous Heart Rate Monitoring and Interpretation	United States	2014-03-13	2013-09-04	U.S. 10,945,677
U.S. Patent App. Pub. No. 2015/0088431 ("Podhajsky")	Dynamic Profiles	United States	2015-03-26	2014-09-19	U.S. 10,945,677
U.S. Patent App. Pub. No. 2015/0371028 ("Patel")	Wearable Electronic Device and Method for Securing Same	United States	2015-12-24	2014-08-21	U.S. 10,945,677
U.S. Patent No. 9,442,523 ("Lee")	User-Wearable Devices with Power Conserving Features	United States	2016-09-13	2014-07-25	U.S. 10,945,677
WO 2015/139980 A1 ("Presura")	Heart Rate Monitor Device	France	2015-09-24	2014-03-14	U.S. 10,945,677
U.S. Patent No.	Printed Wiring	United States	2005-01-11	2002-12-13	U.S. 10,978,789

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
6,841,738 ("Michiwaki")	Board Having Rigid Portion and Flexible Portion, and Method of Fabricating the Board				
U.S. Patent No. 8,705,905 ("Langer")	Printed Circuit Board Element and Method for the Production Thereof	United States	2014-04-22	2007-09-21	U.S. 10,978,789
U.S. Patent No. 9,756,733 ("Drzaic")	Display and Multi-Layer Printed Circuit Board with Shared Flexible Substrate	United States	2017-09-05	2013-04-04	U.S. 10,978,789
U.S. Patent No. 8,383,948 ("Aoyama")	Flex-Rigid Wiring Board and Method for Manufacturing the Same	United States	2013-02-26	2009-09-18	U.S. 10,978,789
U.S. Patent Publication No. 2014/0147128 ("Han")	Structure for Connecting Electrical Trace Lines of Printed Circuit Boards and Optical Transceiver Module with The Same	United States	2014-05-29	2012-11-26	U.S. 10,978,789
U.S. Patent Publication No. 2012/0325524 ("Naganuma")	Flex-Rigid Wiring Board and Method for Manufacturing The Same	United States	2012-12-27	2011-06-23	U.S. 10,978,789
U.S. Patent No. 10,064,276 ("Williams")	3D Bendable Printed Circuit Board with Redundant Interconnections	United States	2018-08-28	2015-10-21	U.S. 10,978,789

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
U.S. Patent Publication No. 2013/083496 (“Franklin”)	Flexible Electronic Devices	United States	2013-04-04	2011-09-30	U.S. 10,978,789
U.S. Patent No. 10,727,568 (“Cho”)	Electronic Device Including High-Frequency Transmission Circuit	United States	2020-07-28	2017-06-13	U.S. 10,978,789
U.S. Patent Application Publication No. 2017/0332479 (“Luo-Larson”)	Multi-Flex Printed Circuit Board For Wearable System	United States	2017-11-16	2016-02-26	U.S. 10,978,789
U.S. Patent Application Publication No. 2019/0045630 (“Kim”)	Flexible Printed Circuit Board	United States	2019-02-07	2016-05-12	U.S. 10,978,789
JP2004247616A (“Sekiguchi”)	Flexible Board	Japan	2003-02-14	2004-09-02	U.S. 10,978,789
U.S. Patent Application Publication No. 2017/0214121 (“Ganchrow”)	Signal Delivery and Antenna Layout Using Flexible Printed Circuit Board (PCB)	United States	2017-01-25	2012-01-26	U.S. 10,978,789
U.S. Patent Application Publication No. 2018/0151951 (“Ikemoto”)	Multilayer Substrate and Electronic Device	United States	2018-05-31	2015-07-30	U.S. 10,978,789
U.S. Patent Application Publication No.	Flexible Interconnect Structure For a	United States	2014-09-18	2013-03-15	U.S. 10,978,789

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Prior Art	Title	Country	Issue/Pub. Date	Priority/Filing	Asserted Patent
2014/0268593 ("DeRosa")	Sensor Assembly				
U.S. Patent Application Publication No. 2012/0206302 ("Ramachandran")	Chassis-Excited Antenna Apparatus and Methods	United States	2012-08-16	2011-02-11	U.S. 10,978,789
U.S. Patent Application No. 2015/0091674 ("Kato")	High-Frequency Transmission Line and Electronic Device	United States	2015-04-02	2013-06-27	U.S. 10,978,789
JP2015-151346A ("Ikemoto '346")	Multilayer Substrate and Electronic Device	Japan	2017-11-16	2015-07-30	U.S. 10,978,789
KR 10-2016- 0022978 ("Kim '978")	Flexible Printed Circuit Board	Korea	2017-09-05	2016-02-26	U.S. 10,978,789
U.S. Patent Application Publication No. 2015/0280771 ("Mow")	Tunable Antenna Systems	United States	2015-10-01	2010-07-06	U.S. 10,978,789

II. Non-Patent Publications

The following table lists prior art non-patent publications which are identified by author, title of the reference, page numbers and publication date. Where individual pages are not provided, Respondents reserve the right to rely on the entire document. Each listed publication became prior art at least as early as the dates provided on the face of the document. Respondents reserve the right to rely upon earlier dates of publication or public availability to the extent such information is uncovered during discovery. Where an asserted patent is identified in relation to a prior art reference, such relationship is exemplary only and does not limit the use of that prior art reference to support the invalidity of any of the asserted patents or to support the state of the art regarding each of the asserted patents.

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Prior Art	Publication Date	Asserted Patent
Dixon et al., “Compressed Sensing System Considerations for ECG and EMG Wireless Biosensors,” IEEE Trans. on Biomed. Circuits and Sys., 6(2):156-166; pages 156-164	2012-04-16	U.S. 10,642,359
Wei et al., “A New Wristband Wearable Sensor Using Adaptive Reduction Filter to Reduce Motion Artifact”; pages 278-81	2008-05-30	U.S. 10,945,677

III. System Art

The following table lists prior art products and systems which are identified by brand name and model name or number of the article and the date of public accessibility, date and location of the offer or sale, and the name and address of any corroborating witness. Respondents reserve the right to rely upon earlier dates of public availability to the extent such information is uncovered during discovery. Where an asserted patent is identified in relation to a prior art reference, such relationship is exemplary only and does not limit the use of that prior art reference to support the invalidity of any of the asserted patents or to support the state of the art regarding each of the asserted patents.

Prior Art Products and Systems	Date	Location	Asserted Patent
Basis B1 Band	Prior to August 20, 2013	United States	U.S. 10,642,359 U.S. 10,231,675
Oura Ring 1	Prior to August 12, 2015	United States	U.S. 10,231,675 U.S. 10,945,677 U.S. 10,978,789
Microsoft Band	Prior to August 12, 2015	United States	U.S. 10,231,675
Apple Watch Series Zero	Prior to August 12, 2015	United States	U.S. 10,231,675
Apple iPhone 6	Prior to June 13, 2016	United States	U.S. 10,978,789

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Samsung Galaxy S5	Prior to June 13, 2016	United States	U.S. 10,978,789
Samsung Galaxy S6	Prior to June 13, 2016	United States	U.S. 10,978,789
Samsung Galaxy S6 Edge	Prior to June 13, 2016	United States	U.S. 10,978,789

CERTIFICATE OF SERVICE

I hereby certify that a copy of the foregoing **RESPONDENTS' NOTICE OF PRIOR ART AS TO U.S. PATENT NOS. 10,642,359, 10,231,675, 10,945,677, AND 10,978,789** was served upon the following as indicated below on May 13, 2026:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, DC 20436	Via EDIS
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W., Room 317 Washington, DC 20436	Via email and Box: JohnsonHines1478@usitc.gov
Counsel for Complainants Samsung Electronics Co., Ltd and Samsung Electronics America, Inc. D. Sean Trainor O'MELVENY & MYERS LLP 1625 Eye Street, NW Washington, D.C. 20006-4061 Telephone: (202) 383-5300	Via email: OMM-SamsungOura@OMM.com

/s/ Jasjit S. Vidwan

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